



sokona^lysis

The Cipher Toolkit Built For All Skill Levels

User Guide

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Introduction

Welcome to sokonalysis v4.0.0 a cryptographic desktop application meant for all skill levels that can be accessed in both Windows and Linux. This application was developed by Soko James and Rosaria Phiri from Kapasa Makasa University featuring a Command Line Interface (CLI) as their Cybersecurity final year project.

Requirements

Linux

The current version works well on all distros including Kali, Parrot and Ubuntu. Ensure to have the following tools installed externally -

1. John The Ripper
2. Aircrack-ng Suite
3. Steghide
4. Stegseek
5. OpenSSL

Windows

The current version works well in Windows 10 and 11. For cracking hashes, John The Ripper as been integrated in sokonalysis and can be used without any errors. For other options requiring external tools can not be utilized in Windows for now. We recommend, you use Linux or wait for coming updates that will integrate those external tools.

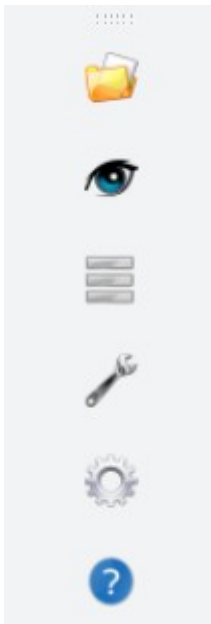
Features

Memory/Cashe

All your preferences such as theme, layout, header position, wordlist and quadgrams paths will be cashed. Your settings will restart where you ended even if you close the app.

Header

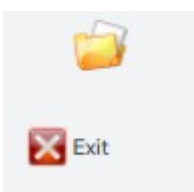
The header is customizable from top to left or right depending on your preference. To achieve this, drag  from the left side of the header to any position of your choice, e.g., left, right or top again.



File

Exit

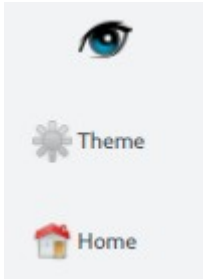
This enables the user to **close** the application.



View

Switch Theme

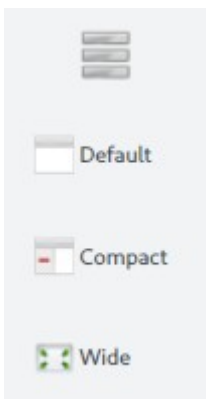
By **default** the applications theme is **light** mode when the applicaion is installed, but will depend on your preference later.



To switch into **dark mode**, you can click on Switch Theme under View or use the shortcut, **Ctrl + Shift + T** on the keyboard or tell the Assistance to help you achieve this.

Layout

When you maximize the window, it will trigger Wide mode, while when minimized it will trigger compact mode.



Default

By default the application opens in the Default layout when you just install the app, then later depends on your preference. This opens the application in a **minimized screen** window mode.

Compact

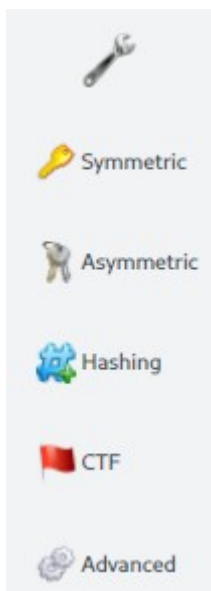
This layout opens the application in a smaller **minimized screen** window mode than the Default layout.

Wide

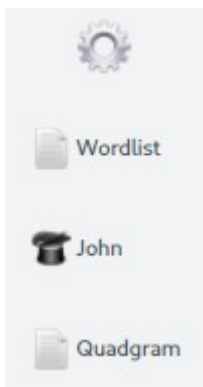
This layout opens the application in **full screen**.

Tools

These are just **shortcuts** of the **main menu**, and can be used when a user is in a category and wants to switch into another category without navigating back multiple times.



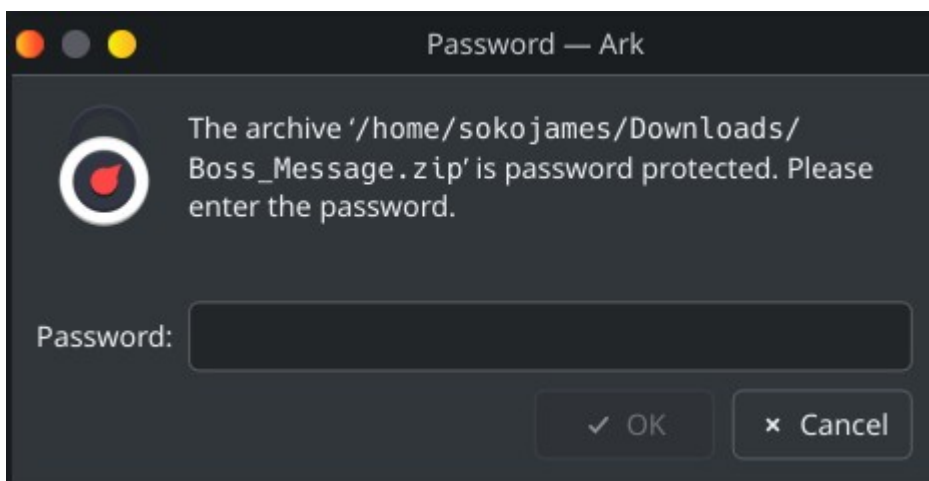
Config



Wordlist Settings

Based on the wordlist used, the system will **slit** it into **multiple parts** based on the system cores or threads, users can also set up the number of parts to be slit. This helps speed up cracking process.

For example, you want to crack a protected documents password, in this case for a CTF challenge on Cyber Talents called Boss Message.



If you are just getting started in CTFs, you can just **drag and drop** the **file** onto the **Assistant**, to get some info.

Results: 1337

Type: Password

Ask Assistant about any tool... (or drop files here)



Notice we have the password. Another was is using the Documents page and select Zip File. Notice we need to **configure** the **wordlist** first

[Back](#) **Crack ZIP File Password**

No Wordlist John Ready

Using the Config tab then Wordlist Settings can help us achieve this. In most cases, the mostly commonly used wordlist is **rockyou.txt** which can be found via **/usr/share/wordlists/** in a Linux system file structure.

Back **Configurations** Wordlist Ready

Configure wordlist settings for all hashing and cracking operations.

Wordlist File

rockyou.txt
133.4 MB

~17,490,188 words • 133.4MB

/home/sokojames/Downloads/rockyou.txt Browse

Wordlist Splitting

☒ Auto-detect optimal split count

Number of parts: 8

Recommended: 8 parts (based on ~17,490,188 words, 8 CPU cores)

Save Configuration

Save the configurations, then drag and drop your zip file then press **Start Cracking**.

Back **Crack ZIP File Password** Wordlist Ready John Ready

Setup Status Results Requirements

1. Cracking Rules (Drag_Drop)

No rules (fastest) Standard rules (recommended) All rules (thorough)

Standard rules (recommend...

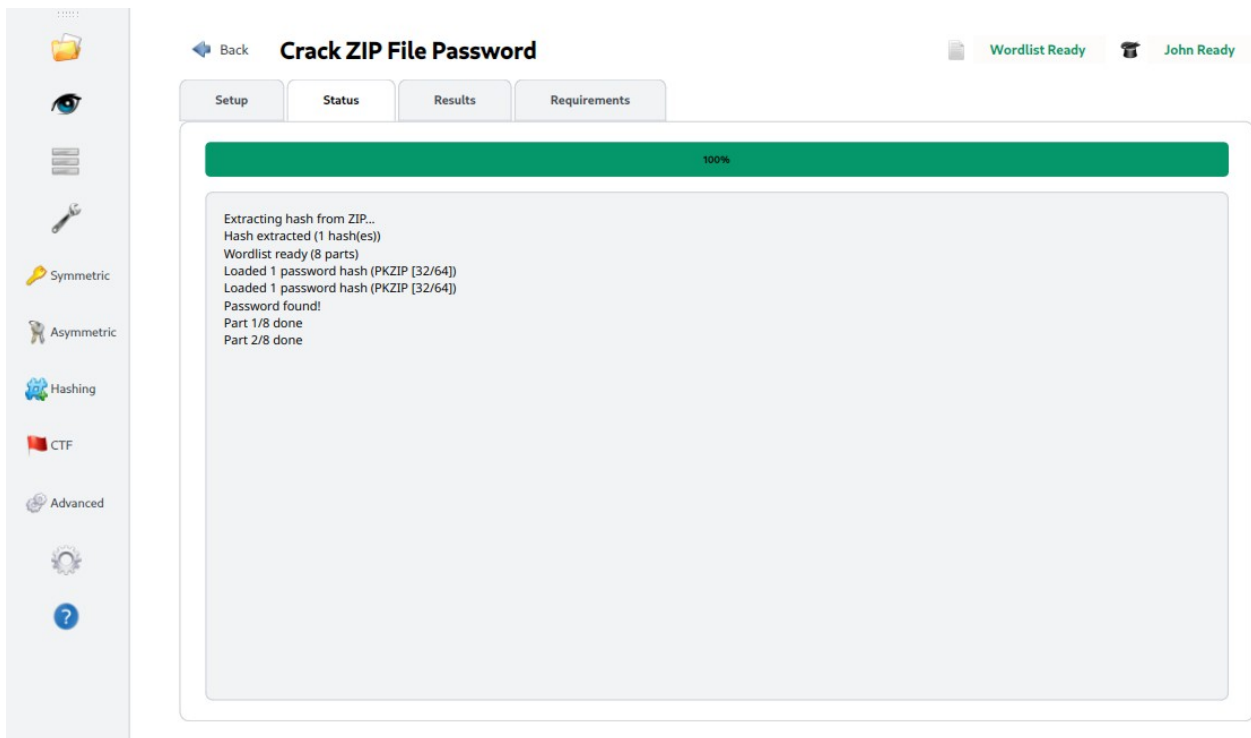
2. ZIP File

Boss_Message.zip
16.9 KB

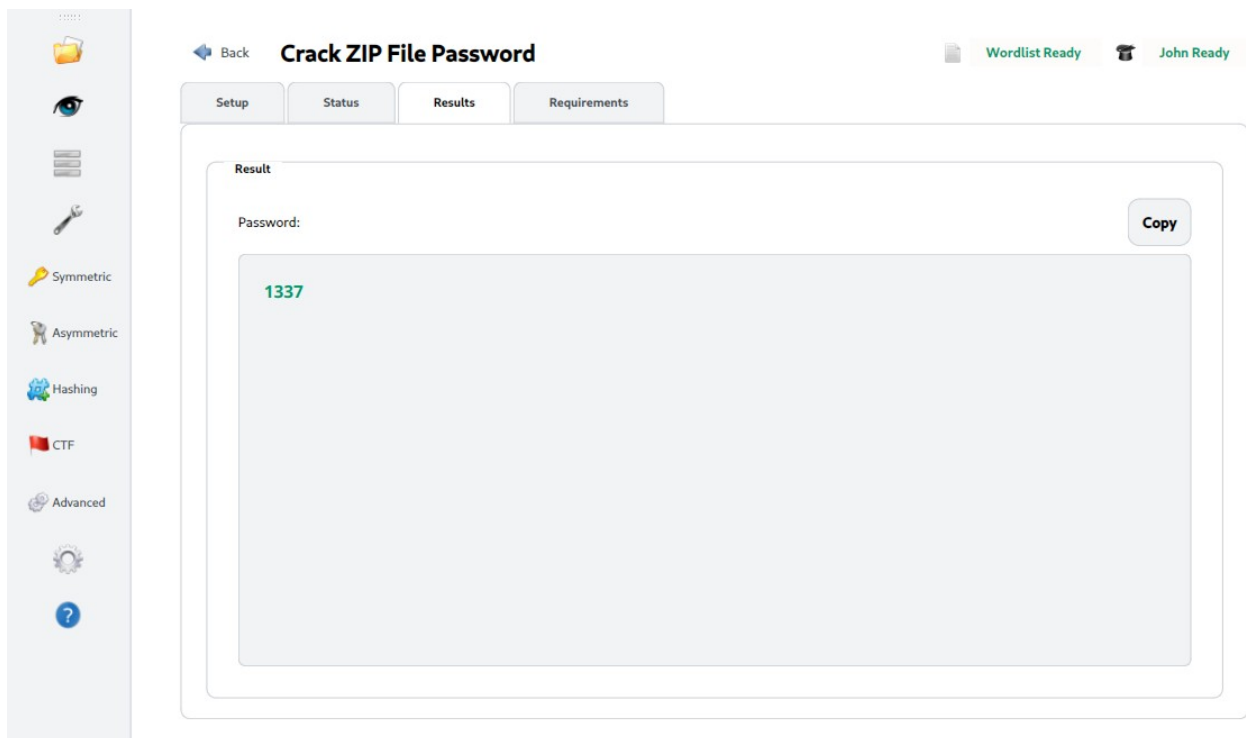
/home/sokojames/Downloads/Boss_Message.zip Browse

Stop Start Cracking

Notice the **wordlist splitting** taking place and the password was found in less than a minute.

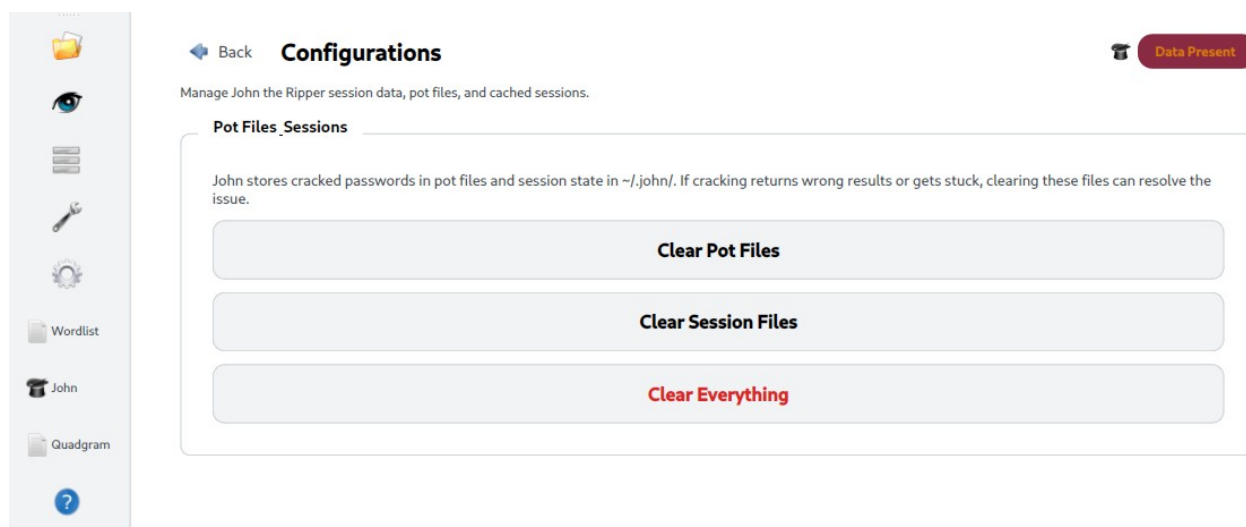


We can use the found password to unzip the Zipped File.



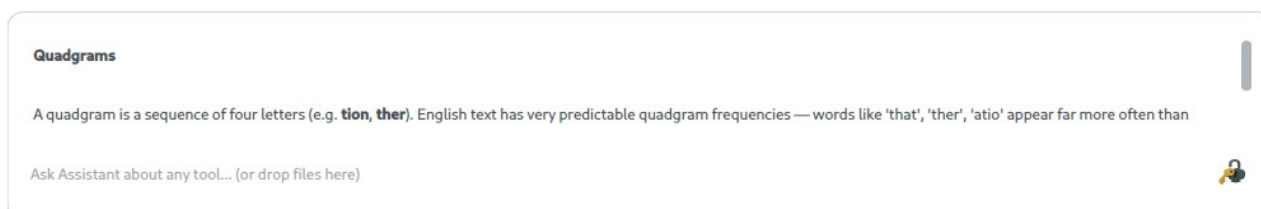
John Settings

Most of the Document password cracking passwords such as Zip File use **John The Ripper**, sometimes john **crashes** its **pot** files where the cracked passwords are saved. So in cases where the john fails to crack a password using a wordlist but the password is in the wordlist, clear the session logs via **Config** the **John Settings**.

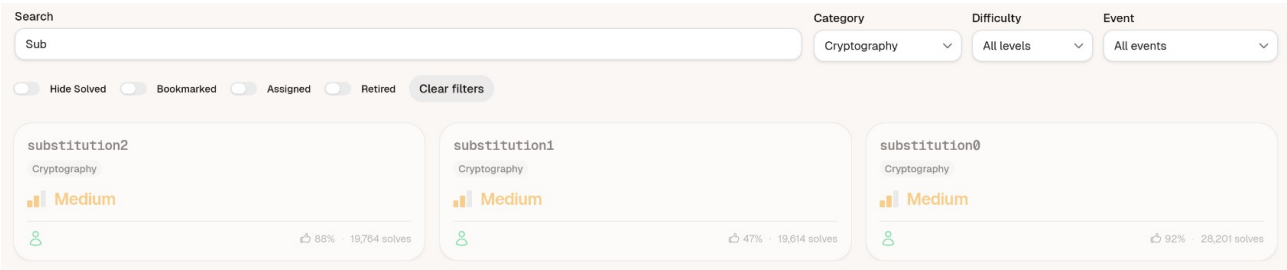


Quadgram Settings

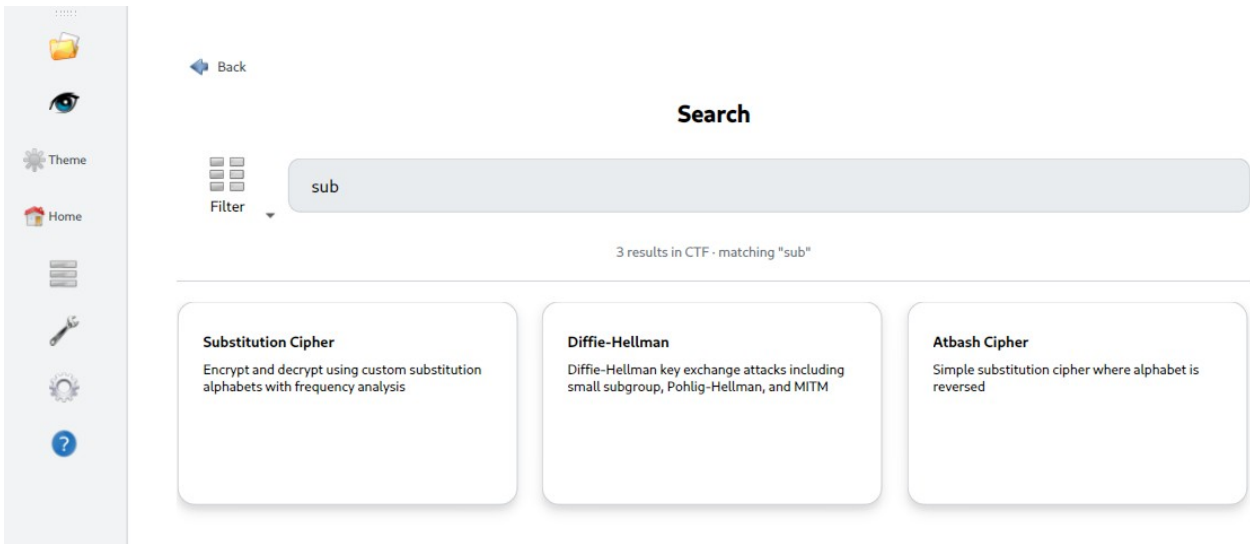
This is useful in a **Substitution Cipher** brute-force cracking using **Frequency Analysis**. The AI assistant can provide us with more info about it.



Let's take an example, on **CyLab Security Academy** famously known as **PicoCTF**, they are multiple **substitution cipher challenges** we can work with.



In this demo, will use **substitution2** challenge. Download the message.txt file to view the cipher text. Using the **Search** option, type sub and **Filter** by the **CTF** category.



Select **Substitution Cipher** and then **Frequency Analysis**. Unlike the wordlist option, quadgrams are not common, use we provide a default **json** file that you can use. Paste the cipher text and press **Break Cipher**.

📁

👁️

⚙️

🔧

🔑 Symmetric

🔑 Asymmetric

🔑 Hashing

🚩 CTF

⚙️ Advanced

⚙️

?

Back

Substitution Cipher - Frequency Analysis

Default (EN.json)

Default

Hill-Climbing Settings

Max Rounds: 10000 Consolidate: 3

Input

Status

Results

Ciphertext

xcslhlguxulnlsrmpxcisklmmmluxrwmguclgctuecpmpmepjyvxlsulevsgxbepjylxgxpauaemvogatebwlsvyrxsgpxraovuebwlfecrmmlatxlclulepjylxgxpaupevuyvgj
rsgmbpaubxljurojgaguxsrxgpaivaorjlxarmukcgecrslnlsbulvmraojrszlxrwmuzgmmucpklnlsklwlmglnclyspylsvysypulpirctuecpmpmepjyvxlsulevsgxbepjylx
gxpagaupxpxmbxpxlrecnrmvrwmluzgmmuwxxrmupxtlxuxvolaxugaxlsxllogaraolhegxlwpxvpepjyvxlsueglaelolilaugnlepjylxgxpauurslpixlamrwpsgpvurir
gsuraopjlopkaixpsaagateclezmguxuraolhlevxgatepaigtuesyxiupilaupaxclpxclsraogucrlngmbipevulopalhymprxgparaogjxpgnurgxparaopixlacrulmjljaxu
piymrbklwlmglnclyspylxgxpaxpvegatpaxclpiilaugnllmjlaxupiepjyvxlsulevsgxbguxclsipsrlwxxslncgemplsxlclnratimgujxpxvolaxugarljsgercgtuecpmpm
vscisklwmglnclyspylxgxpaxpvegatpaxclpiilaugnllmjlaxupiepjyvxlsulevsgxbguxclsipsrlwxxslncgemplsxlclnratimgujxpxvolaxugarljsgercgtuecpmpm
nlepjylxgxpauopluapxmxlrouxvolaxupzapkxclqslaljlbrulillexgnlmbxrlrecgatxcljprexgnlmbxcgqazmgzlarxxrezlsreroljbgurapiilaugnmbpsqlaxlocgtuecpmpm

Paste Clear

Configure custom quadgram JSON via Config > Quadgram Settings.

Break Cipher

There we go, we have our **flag**

📁

👁️

⚙️

🔧

🔑 Symmetric

🔑 Asymmetric

🔑 Hashing

🚩 CTF

⚙️ Advanced

⚙️

?

Back

Substitution Cipher - Frequency Analysis

Default (EN.json)

Default

Hill-Climbing Settings

Max Rounds: 10000 Consolidate: 3

Input

Status

Results

Decryption Results

Recovered Key:

Copy Key

Alphabet: abcdefghijklmnopqrstuvwxyz Key: rweolitegdzmjapjfsuxvknhbq

Decrypted Text:

Copy Text

and improvisation and or tennis elements or play we believe a competition touching on the offensive elements or computers
security is therefore a better vehicle for the evangelism of most students in American high schools further we believe that an under
standing of offensive techniques is essential for mounting an effective defense and that the tools and configuration focus on
countered in defensive competitions does not lead students to know their enemy as effectively as teaching them to actively thi
nk like an attacker academy is an offensively oriented high school computer security competition that seeks to generate inte
rest in computer science among high schoolers teaching them enough about computer security to pique their curiosity moti
vating them to explore on their own and enabling them to better defend their machines the flag is academy{N6R4M_4N41Y51
5_15_73D10U5_FD56C357}

Similar to the Wordlist Settings, we can also **configure** a **custom** quadgram json file. First, switch from **Default (EN.json)** to **Custom** **Json**.

Substitution Cipher - Frequency Analysis

Custom JSON

Back

Hill-Climbing Settings

Max Rounds: 10000 Consolidate: 3

Input Status Results

Decryption Results

Recovered Key: Copy Key

Alphabet: abcdefghijklmnopqrstuvwxyz Key: rweolitcgdzmjapysuxvknkbbq

Decrypted Text: Copy Text

and improvisation and ortennaselements or play we believe a competition touching on the offensive elements or computer security is therefore a better vehicle for the evangelism to students in American high schools further we believe that an understanding of offensive techniques is essential for mounting an effective defense and that the tools and configuration focus encountered in defensive competitions does not lead students to know their enemy as effectively as teaching them to actively think like an attacker academy is an offensively oriented high school computer security competition that seeks to generate interest in computer science among high schoolers teaching them enough about computer security to pique their curiosity motivating them to explore on their own and enabling them to better defend their machines the flag is academy{N6R4M_4N41Y515_15_73D10U5_FD56C357}

Under **Config** select **Quadgram Settings** then **drag and drop** your custom **json** file.

Configurations

JSON Ready

Back

Configure the quadgram JSON file used for substitution cipher frequency analysis.

Quadgram JSON File

EN.json

3.4 MB | Alphabet: 'abcdefghijklmnopqrstuvwxyz' (26 chars)

Valid quadgram file | Alphabet: 'abcdefghijklmnopqrstuvwxyz' (26 chars) | Most frequent: 'tion' | Total: 93,609,337

/home/sokojames/Desktop/Cyber Security/Projects/sokonalysis/sokonalysis/gui/EN.json

Browse

Save Configuration